

POWER BI ADVANCED SERIES

DAY 9: INTERACTIVE DASHBOARD DEVELOPMENT

Transition from static report visualization architectures to professional, exploratory business intelligence assets.

 **2 Hours**
Live Framework

 **Business Ready**
Exploratory Analytics

Executive Framework

The shift from consumption to interactive discovery

Until now, we cleaned data, structurally transformed arrays, and authored analytical DAX queries."

It today — we make dashboards think and adapt dynamically.

Modern corporate end-users do not merely scan hardcoded reports; they require analytical workflows that allow them to probe cross-unctional relationships in real time.

Session Learning Targets

- ✓ Deep Hierarchy Navigation (Drill-Down)
- ✓ Target-Context Redirection (Drill-Through)
- ✓ UI State Preservation (Bookmarks & Toggles)
- ✓ Contextual Micro-Dashboards (Custom Tooltips)

1. Drill-Down Hierarchy Engine

Granular progression inside unified visual coordinates

Definition: Allows users to navigate progressively deeper inside an explicit analytical schema without switching windows.

STRUCTURAL ANALOGY

Geospatial Resolution: Global Country View → Regional State Boundary → Metropolitan Coordinates → Street address mapping profiles.

Academic Hierarchy

Branch → Year → Semester

Enterprise HR Hierarchy

Department → Role → Employee

Implementation Protocol

- 1 Right-click base field in Data pane → **Create Hierarchy**
- 2 Add sub-dimensional attributes sequentially into hierarchy stack
- 3 Bind newly created hierarchy block to Visual X-Axis
- 4 Activate the contextual native ****Drill Mode Toggle**** (Down Arrow)

2. Contextual Drill-Through Actions

Targeted multi-page record parsing across data boundaries

Configuration Blueprint

- 1 Initialize dedicated target workspace page (e.g., "Student/Employee Profile")
- 2 Locate the **Drill-through field bucket** within the Page Format parameters
- 3 Map unique identifier key (e.g., **RollNo** / **EmployeeID**) into target well
- 4 Right-click high-level dimensional row on landing page → Select target route

Definition: Transitions the analyst to an entirely isolated detail sub-page while instantly carrying over the selected context filter parameters.

OPERATIONAL FLOW

High-level performance monitoring dashboard → Right click specific outlier point → Launch hyper-focused context sheet presenting metrics unique to that entity.

```
// Power BI Context Isolation Filter Engine  
SELECTED_CONTEXT: Page1.Visual.SelectedValue(RollNo =  
"R101") → Pass to Page2.Filters
```

3. UI State Bookmarks & View Toggles

Capturing spatial metadata states for interactive storytelling

Definition: Serializes and saves the active view configuration of a report page—including filter context, visual visibility states, and spatial positioning matrix parameters.

ANALOGY FRAME

Streaming Context States: Like a checkpoint save state within video game engines or streaming video bookmarks—instantly restore exact parameters precisely where operations left off.

Core Enterprise Use Case

Single Screen Toggle Real Estate: Seamlessly switch canvas footprint layout between a performance review state and an operational cost matrix without increasing presentation tab overhead.

State Machinery Set-Up

- 1 Activate native **Selection Pane** and **Bookmark Pane** options via View Tab
- 2 Manually configure UI components to desired primary display visibility layout
- 3 Capture snapshot via Bookmark Pane → Label accordingly (e.g., "CGPA View")
- 4 Invert UI component visibility settings via Selection menu → Save as alternative Bookmark

4. Action Buttons & Native Navigation Links

INTERFACE LAYOUT

Constructing web-application mechanics directly within BI reports

Button Mapping Strategy

- 1 Ribbon Selection: Navigate to **Insert** menu → Deploy **Buttons** element
- 2 Select **Blank Canvas Button** construct or programmatic Page Navigator block
- 3 In Formatting Card: Toggle **Action Controller** property parameter to **ON**
- 4 Assign explicit Type definition: **Page Navigation** or link directly to custom **Bookmark**

Definition: Instantiates physical interactive user touchpoints to map execution vectors across discrete layout views or destination sheets.



Page Navigation Matrix

Overview → Analysis → Fin Ledger



Bookmark Integrations

Instantly swap state via user click interface

Architectural Standard: Ensure consistency in the placement of control panel so users can intuitively navigate between reports without unexpected shifts in layout structure.

5. Custom Contextual Tooltips

Constructing dynamic context menus inside real-time canvas layers

Definition: Repurposes entire operational canvas pages into reactive micro-insight windows that trigger upon tracking coordinates over analytical visual dimensions.

CONSUMER UX ANALOGY

Modern e-commerce platform matrix interfaces: Hovering over tracking points surfaces micro-graph modules mapping delivery telemetry without changing the current view context.

Instead of traditional string text descriptors, custom tooltips display fully formatted mini charts, performance indicators, or identity details specific to the selected data point.

Tooltip Canvas Provisioning

- 1 Initialize isolated workspace sheet → Set **Page Information** to **Tooltip = ON**
- 2 Under Canvas Settings → Adjust specialized scaling geometry size option to **Tooltip**
- 3 Populate macro workspace frame with clean summary indicators and charts
- 4 On main dashboard data visual: Map format rules → Tooltip options → Report Page destination

Live Lab Environments: Datasets Schema

ARCHITECTURE

Structural breakdown of deployment assets

Academic Operations Data Package

1. Student_Overview.csv (400 records)

Keys: RollNo, StudentName, Branch, Year, Sem, CGPA, Attendance, PendingFee

2. Student_Fee_Detail.csv (400 records)

Keys: RollNo, CollegeFee, PendingFee, ScholarshipStatus

3. Performance_Detail.csv (500 records)

Keys: RollNo, Branch, Subject, Marks, Semester

Enterprise HR Operations Data Package

1. HR_Overview.csv (450 records)

Keys: EmployeeID, Department, Role, Salary, Performance, Attendance

2. HR_Salary_Detail.csv (500 records)

Keys: EmployeeID, Month, Salary, Bonus

3. HR_Department_Detail.csv (500 records)

Keys: Department, Role, EmployeeID, PerformanceScore

20-Minute Production Sprint

SPRINT CHALLENGE

Deploying real-world feature parameters under execution deadlines

Target Path A: Academic Deployment

Construct a functional Academic Command Center featuring:

- Hierarchical analysis drilling from Branch level down to Semester details.
- A Drill-Through view for comprehensive student profile lookups.
- A single-click canvas UI toggle interface to switch view context between Tuition Fees and Grade point tracking modules.

Target Path B: Enterprise HR Operations

Construct an Executive Workforce Management Board featuring:

- Structural drills navigating Department heads directly through discrete operational individual contributors.
- An Employee Detail context card utilizing dedicated drill-through architecture.
- Hover tooltips surfacing multi-series historical cost trends immediately on high-level charts.

Interactive Review & Common Pitfalls

Evaluating understanding of analytical context patterns

Drill-Down vs Drill-Through?

Drill-Down handles hierarchical unpacking inside the same visual window space. **Drill-Through** updates the filtering context across entirely different destination sheets.

Why use Bookmarks?

They allow you to present structured analytical stories. Instead of shifting across multiple screens, you present layered data perspectives using clean view configurations on a single canvas.

Why Custom Tooltips?

They provide condensed, on-demand insights. By designing localized hover cards, you keep high-level report structures clear while offering deep contextual detail right when it's needed.

Data Architecture Rule

"Good dashboards display clear summaries; great dashboards give users the interactive controls to discover insights on their own."

